# Andrew Li

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## Education

**University of Waterloo** — Waterloo, ON
*Bachelor of Computer Science (Honours, Co-op)* — *Class of 2031*

## Projects

**Nebulosity** | *TypeScript, MCP, SQLite, Next.js* | nebulosity.ca | Demo Video — Jul. 2026 – Oct. 2026

- Built an MCP server and code index that made Claude Code up to 4.4× faster on questions that span many files, with all 42 answers fully correct versus 29 without it
- Wrote an indexer for 7 languages that maps all 14,849 files and 140,000+ imports in VS Code's codebase into SQLite and updates on every save, shown as a live map in a published VS Code extension
- Evaluated it over 378 Claude Code runs on 3 open-source codebases, checking every answer against independent reference answers, including one it was never tuned for (up to 3.7× faster)

**OlympIQ** | *Next.js, TypeScript, Firebase, Firestore* | olympiq.ca — May 2026 – Jul. 2026

- Built a scheduling platform for a 60+ member Science Olympics team, replacing 10+ hours of manual spreadsheet work with a custom algorithm that assigns competitors to their preferred events while avoiding time conflicts
- Shipped it solo and piloted it with the executives before the full team adopts it in the 2026–27 season

**Suplex** | *Python, FastAPI, Shapely, three.js, React* — Hack the Valley 2026

- Led the technical design of a tool that checks any Toronto address against the City's 4 pre-approved multiplex designs and shows the result in 3D
- Built the rules engine, which screened all 495,456 Toronto lots against the zoning by-law and answers a single address in under 50 ms, plus the FastAPI backend and three.js 3D scene

## Experience

**Software Engineer** — Sep. 2026 – Present
*Marble Investments* — *Waterloo, ON*

- Building a news pipeline for the firm's equity research platform that sources, filters, and merges the latest news for every tracked company, with an LLM flagging industry news likely to affect them

**Software Engineer, Ground Station** — Sep. 2026 – Present
*UW Orbital (University of Waterloo Satellite Design Team)* — *Waterloo, ON*

- Developing the Mission Control Centre, the ground station web app operators use to send commands to the team's 3U CubeSat and monitor its telemetry
- Building the operator dashboard that combines the satellite's status, command history, telemetry, and latest captured image into a single interface

**Head Trainer** — Feb. 2023 – Jun. 2026
*Science Olympics Team, Victoria Park C.I.* — *Toronto, ON*

- Led 6 trainers for a 60+ member team, and all 4 junior teams I coached placed 1st at Western University's Science Olympics

## Awards

**Hack the North 2026 (349 projects)**: Best Mobile Experience (Expo), $1,000+ in team prizes
**Hack Canada 2025 (89 projects)**: Best Use of Gemini AI
**Canadian Computing Competition (Senior)**: Distinction, 2024, 2025, 2026
**Competitive Programming**: DMOJ Top 2%, 500+ problems solved
**CALICO Informatics Competition (UC Berkeley)**: Bronze Brick Winner, 2024
**St. Lawrence Coding Competition**: 1st Place Team, 2024

## Technical Skills

**Languages**: C++, Java, Python, TypeScript, JavaScript
**Frameworks**: React, Next.js, Tailwind CSS, three.js, Node.js, FastAPI, Spring Boot
**Cloud & Tools**: Firebase, Supabase, SQLite, Google Cloud, Cloudflare, Vercel, Git, MCP, Claude Code